

Discipline Specific Elective – (DSE) : B.Sc (H) Comp Sc./B.A (Prog)/ B.Sc (Prog)
Sem VII
NEP-UGCF-2022
Digital Image Processing

S.No .	Topic	Reference[1]	No. of Lectures
1	Introduction: Digital Image Fundamentals Brightness, Adaptation and Discrimination, Light and Electromagnetic Spectrum, Image Sampling and Quantization, Some Basic Relationships Between Pixels, Types of images	Ch 1 1.1 (pp -18-19),1.4 (pp 41-43) Ch 2 2.1(pp 50-54) 2.2(pp 54-57, 61-63) 2.4(pp 63-77) 2.5(pp 79-83)	5
2	Spatial Domain Filtering Some Basic Intensity Transformation Functions, Histogram Equalization, Spatial Correlation and Convolution, Smoothing Spatial Filters: Low pass filters, Order Statistics filters, Sharpening Spatial Filters: Laplacian filter	Ch 3 3.1,3.2,3.3 (133-140), 3.4(pp 153-160), 3.5, 3.6(pp 175-182)	10
3	Filtering in Frequency Domain The Discrete Fourier Transformation (DFT), Frequency Domain Filtering: Ideal and Butterworth Low pass and High pass filters	Ch 4 4.4, 4.5(240) 4.7(pp 260-268), 4.8, 4.9(pp 284-289)	5
4	Image Restoration: Noise models, Noise Restoration Filters	Ch 5 5.1(pp 318), 5.2(pp 318-325), 5.3(pp 327- 333)	5
5	Image Compression: Fundamentals of Image Compression, Huffman Coding, Run Length Coding	Ch 8 8.1 (pp 540-553), 8.2, 8.6(pp 566-567)	5
6	Morphological Image Processing: Erosion, Dilation, Opening, Closing, Hit-or-Miss Transformation, Basic Morphological Algorithms.	Ch 9 9.1-9.5 (pp 652-661)	10
7	Image Segmentation: Point, Line and Edge Detection, Thresholding	Ch 10 10.1, 10.2(pp 701-723, 735-736), 10.3(742-746)	5

References :

1. Gonzalez, R. C., & Woods, R. E. Digital Image Processing. 4th edition. Pearson Education, 2017

Additional Resources

1. Castleman, K. R. Digital Image Processing. 1st edition. Pearson Education, 2007
2. Gonzalez, R. C., Woods, R. E., & Eddins, S. Digital Image Processing using MATLAB. Pearson Education Inc., 2004
3. Jain, A. K. Fundamentals of Digital Image Processing. 1st edition Prentice Hall of India, 1988.

Practical List

The Practical are to be conducted using Python/ Scilab. Become familiar with basic Python libraries for Image Processing like OpenCV, Scikit-Image etc. / Scilab Commands.

1. Perform the following:
 - a. Read and display image.
 - b. Resize a given image.
 - c. Convert a given color image into a corresponding gray-scale image.
 - d. Convert a given color/gray-scale image into black & white image f. Draw the image profile.
 - e. Separate a given color image into three R G & B planes.
 - f. Create a color image using separate three R, G and B planes.
 - g. Write given 2-D data in image file.
2. To write and execute image processing programs using point processing method:
 - a. Obtain Negative image
 - b. Obtain Flip image
 - c. Thresholding
 - d. Contrast stretching
3. To write and execute programs for image arithmetic operations:
 - a. Addition of two images
 - b. Subtract one image from other image
 - c. Calculate mean value of image
 - d. Different Brightness by changing mean value
4. To write and execute programs for image logical operations:
 - a. AND operation between two images
 - b. OR operation between two images
 - c. Calculate intersection of two images

- d. Water Marking using X-OR operation
- e. NOT operation (Negative image)

5. To write and execute a program for histogram calculation and equalization:

- a. Using inbuilt function
- b. Without using inbuilt function

6. To write and execute a program performing the following geometric transformations on an image:

- a. Translation
- b. Scaling
- c. Rotation
- d. Shrinking
- e. Zooming

7. To understand various image noise models and to write programs for:

- a. Image restoration
- b. Remove Salt and Pepper Noise
- c. Minimize Gaussian noise
- d. Median filter and Weiner filter

8. Write and execute programs to remove noise from images using spatial filtering.

- a. Understand 1-D and 2-D convolution process
- b. Use 3x3 Mask for low pass filter and high pass filter

9. Write and execute programs for image frequency domain filtering.

- a. Apply FFT on given image
- b. Perform low pass and high pass filtering in frequency domain
- c. Apply IFFT to reconstruct image

10. Write and execute a program for edge detection using different edge detection mask.

11. Write and execute a program for image morphological operations erosion and dilation.